ACS MILESTONE 8

# User Guide

This guide explains how to run the current application, play the sample adventure, save progress, use the editor, validate drafts, and move from a local draft to a published release.

### What is included right now

- The playable runtime at `apps/web/index.html`
- The browser editor at `apps/web/editor.html`
- The local project, validation, and release API at `apps/api/dist/index.js`

# Starting The Application

Build the workspace if needed, then start both servers from the repo root.

### Web Server

```
node .\apps\web\server.mjs
```

Default URL:

```
http://localhost:4173/
```

Common alternate URL in this environment:

```
http://localhost:4317/
```

### API Server

```
node .\apps\api\dist\index.js
```

Local API check:

```
http://localhost:4318/api/session
```

### Editor URL

```
http://localhost:4317/apps/web/editor.html
```

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# Playing The Game

The runtime can load the built-in sample adventure, a local draft playtest, or a published release.

### Sample objective

1. Speak to the Oracle.
2. Enter the shrine.
3. Claim the Solar Seal.
4. Return to the Oracle.

### Keyboard controls

- `WASD` or arrow keys move
- `E` interacts
- `Q` inspects
- `Enter` or `Space` advances dialogue

### Runtime panels

- Map and player position
- Session source
- Objective panel
- Save controls
- State summary and event log

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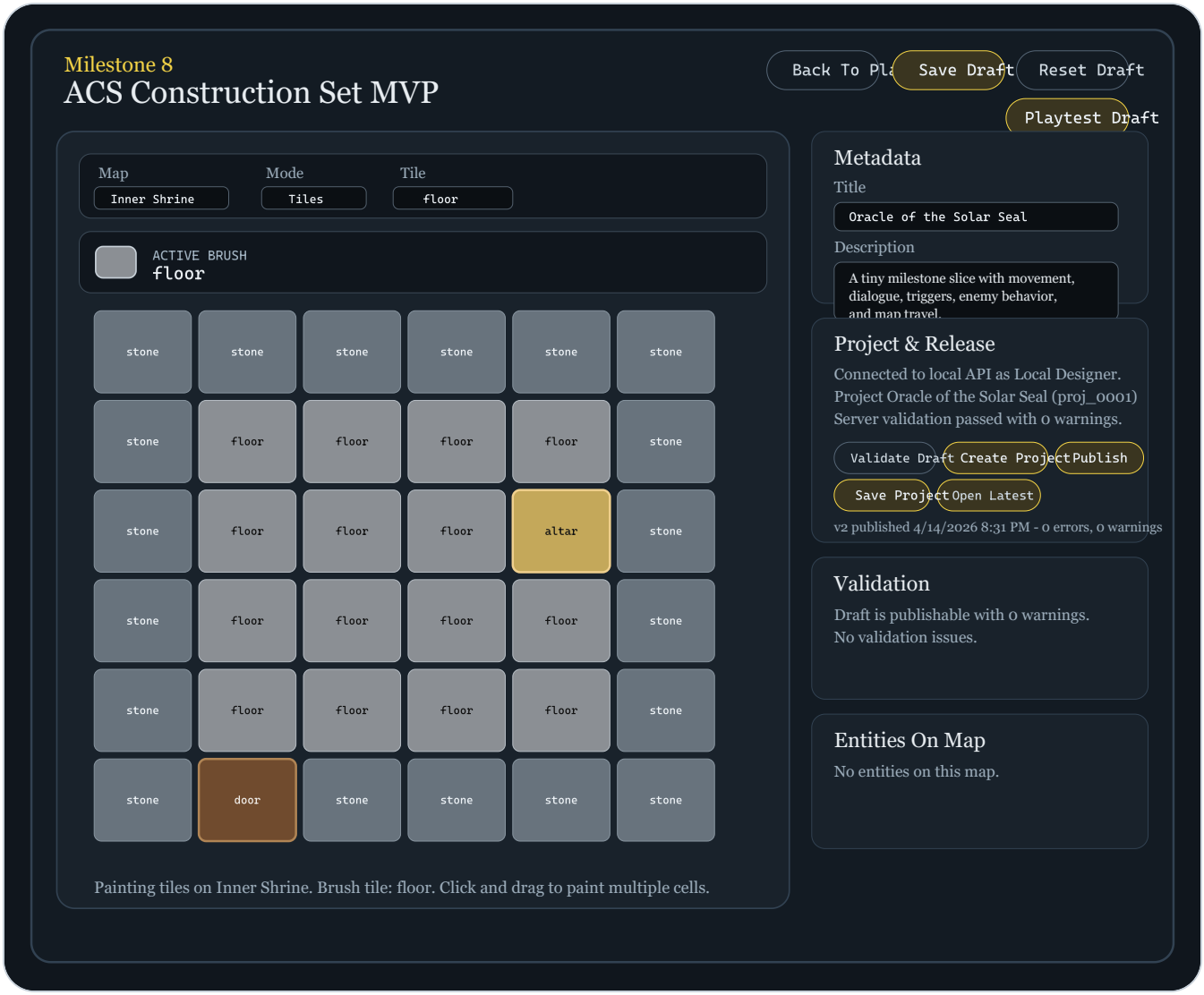
## Saving And Loading Progress

The runtime uses separate local save slots for the sample adventure, draft playtests, and published releases.

- **Save** stores the current runtime snapshot in IndexedDB.
- **Load** restores the most recent snapshot for the current source.
- **Reset** restarts the live session without deleting the saved snapshot.

# Using The Editor

Open the editor at `http://localhost:4317/apps/web/editor.html`.




The current editor supports metadata editing, tile painting with a persistent brush, repositioning existing entities, shared validation summaries, API-side `Validate Draft` checks, local draft saves, backend project saves, publishing releases, and opening those releases in the runtime.

**Top buttons**

- `Back To Play`
- `Save Draft`
- `Reset Draft`
- `Playtest Draft`

**Toolbar controls**

- `Map` chooses the current map
- `Mode` switches between Tiles and Entities
- `Tile` picks the tile to paint
- `Entity` picks the existing entity to move
- `Active Brush` shows the currently loaded tile brush in tile mode

## Tile Editing

1. Choose `Tiles` mode.
2. Select a tile id from the dropdown.
3. Check the `Active Brush` preview to confirm the selected tile.
4. Click a cell in the grid to paint a single tile, or click and drag to paint across multiple cells.

The selected tile stays loaded like a brush until you choose a different tile.

## Entity Editing

1. Choose `Entities` mode.
2. Select an existing entity instance.
3. Click the destination cell.

Current limitation: the editor can move existing entities, but it does not yet create or delete them.

## Validation

The validation panel runs the shared validation package and shows an error-and-warning summary plus a detailed issue list for the current draft. Use `Validate Draft` in the `Project & Release` panel when you want the local API to run the same validation report the publish flow uses. Blocking errors keep project save and publish actions disabled until they are fixed.

# Tutorial: Make A Simple Map Edit And Play It

Because the current editor cannot create a brand-new map yet, the easiest useful tutorial is to modify one of the built-in maps and then playtest the edited version.

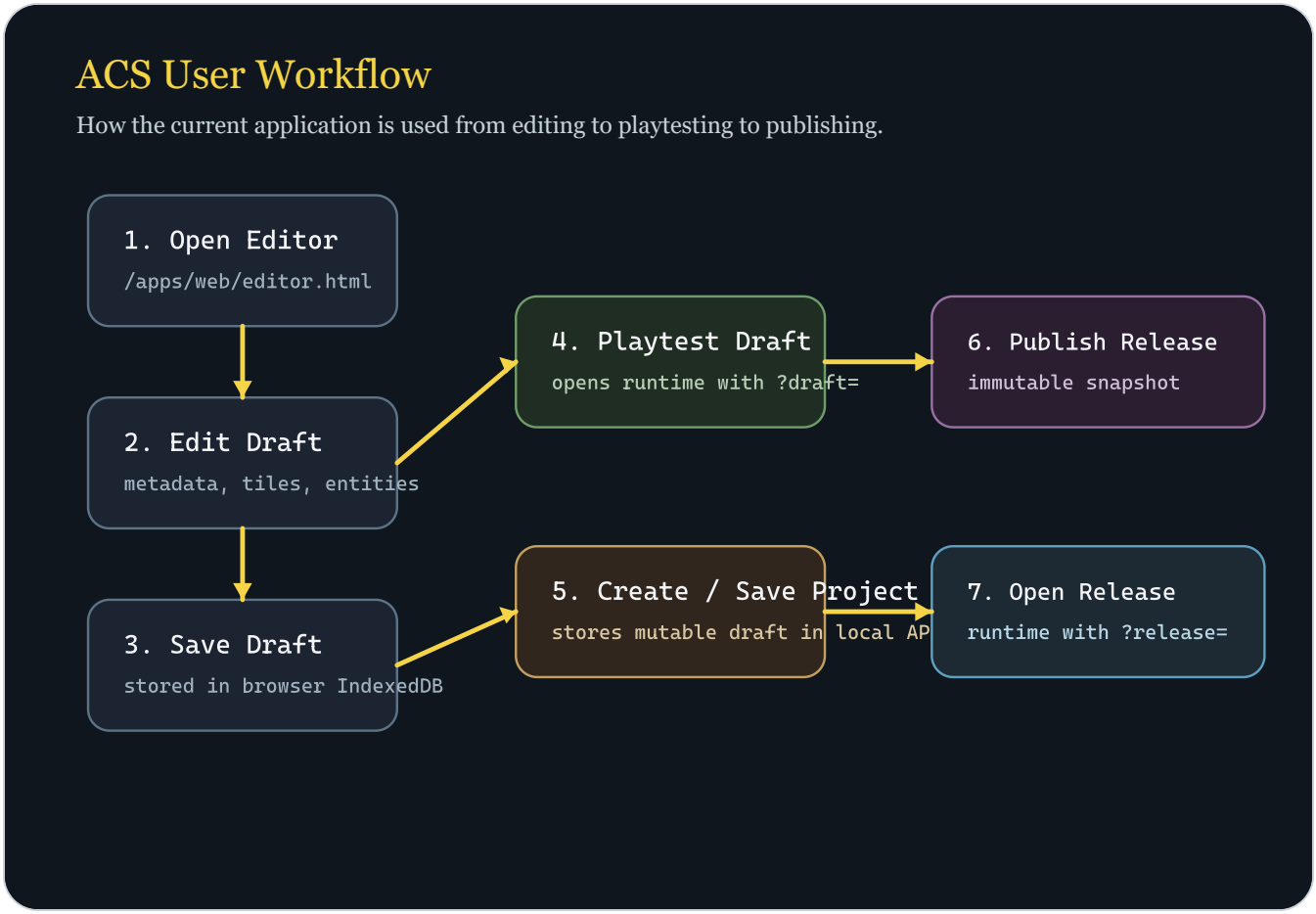
**Goal of this tutorial**

We will edit `Inner Shrine`, paint a few tiles to make the room feel different, validate the draft, save it, and open that draft in the runtime.

1. Open the editor at `http://localhost:4317/apps/web/editor.html`.
2. Choose `Inner Shrine` in the `Map` dropdown.
3. Set `Mode` to `Tiles`.
4. Select `floor` in the `Tile` dropdown.
5. Use the selected tile like a brush. Click individual cells or hold the mouse button down and drag across multiple cells to paint continuously.
6. Leave the `door` tile in place so the map exit still works.
7. Leave the `altar` tile in place if you still want the shrine reward trigger to work.
8. Click `Validate Draft` and confirm the draft is publishable.
9. Click `Save Draft`.
10. Click `Playtest Draft`.
11. In the new runtime tab, walk to the shrine and confirm that the room layout matches your edits.

If the local API is running, you can continue by clicking `Create Project`, `Save Project`, `Publish Release`, and then `Open Latest Release`.

# Projects And Published Releases




The editor can move a draft through five project stages:

1. **Validate Draft**: run the backend validation report without publishing.
2. **Create Project**: create a mutable backend project from the current draft.
3. **Save Project**: update the mutable backend draft.
4. **Publish Release**: create an immutable release snapshot.
5. **Open Latest Release**: launch that published release in the runtime.

### Important distinction

- `Save Draft` writes to browser storage.
- `Save Project` writes to the local API.
- `Publish Release` freezes a release snapshot rather than editing it in place.

## Where Data Lives

**Browser storage**

- Runtime saves
- Local drafts
- Remembered active project id

**Local API storage**

- Mutable project drafts
- Immutable published releases
- Stored locally in `apps/api/data/store.json`

## Current Limitations

- No real user accounts yet
- No cloud backend yet
- No asset upload flow yet
- No creation or deletion of maps in the editor yet
- No creation or deletion of entity instances in the editor yet
- No trigger editor or dialogue editor yet
- The runtime still uses simple colored tiles and abstract markers

## Troubleshooting

### The editor says the API is unavailable

```
node .\apps\api\dist\index.js
```

### Validate Draft fails or publish stays disabled

- A trigger may reference a missing map, item, dialogue, or quest.
- An entity or start position may be outside a map's bounds.
- A tile layer may have the wrong tile count for its dimensions.

### A published release will not open

- The API may not be running.
- The release id may not exist anymore.
- The local API store may have been cleared.

### My draft changes are gone

- Use `Save Draft` for browser-local storage.
- Use `Save Project` for backend storage.

### Playtest Draft opens the sample adventure instead

The runtime falls back to the built-in sample when it cannot find the draft key passed by the editor.